

HEALTHY BITE

Advanced Entity Relationship Diagram (ERD) & Database Schema Design

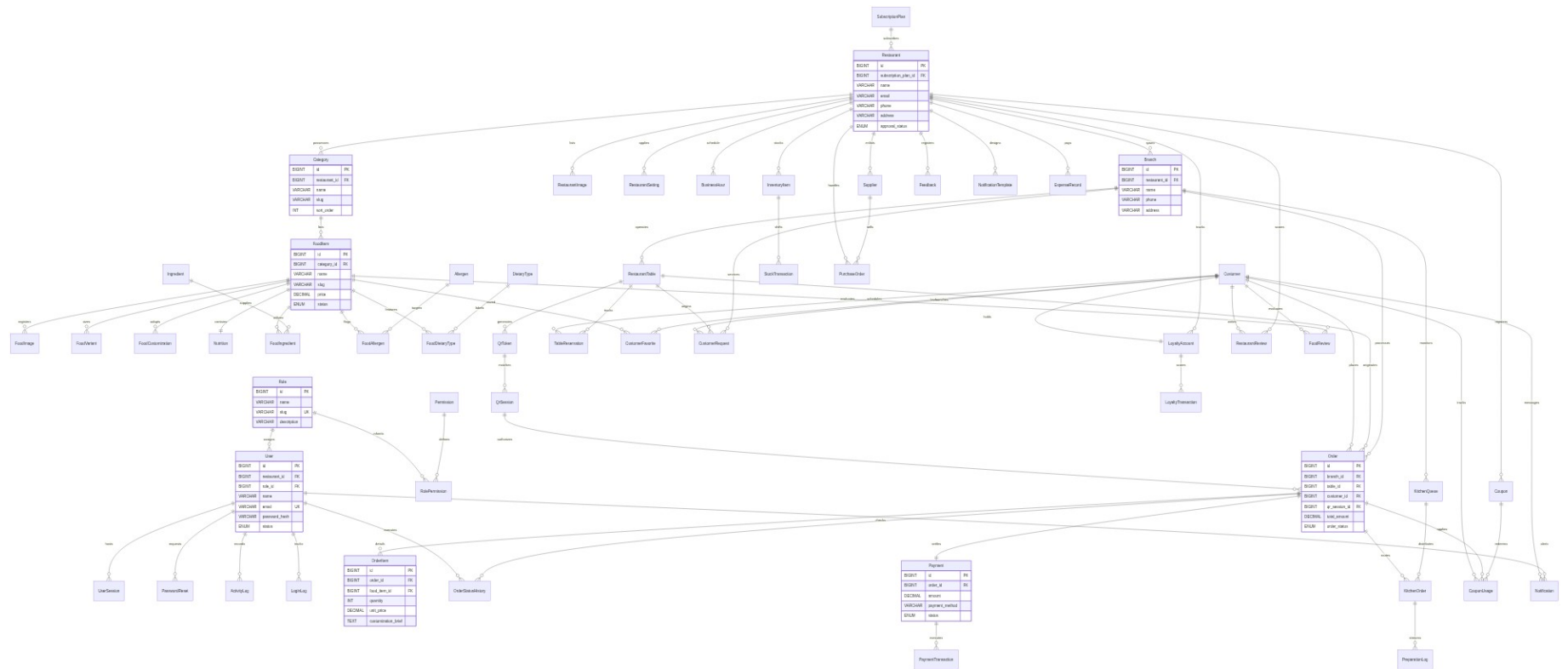
This document details the enterprise-grade, multi-tenant database design for the **Healthy Bite** digital menu and table-ordering platform. Structured in Third Normal Form (3NF), it preserves strict referential integrity under ACID compliance and supports multi-branch scaling, cryptographic table ordering sessions, real-time kitchen routing, and predictive inventory logging.

Project Title:	Healthy Bite QR-based Table Ordering MVP
Academic Context:	BCA Semester V Database Design Submission
System Architecture:	Custom PHP MVC MVC Foundation / MySQL 8.0+
Target Environment:	XAMPP Local Server Integration (Apache & InnoDB Engine)
Version / Date:	v1.0.0 / July 13, 2026

1. Complete ER Diagram

This diagram was created using standard Mermaid.js relational notation.

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2. Entity List

The database schema implements 45 structured tables.

Module A: Authentication

1. **Role**: Defines user roles within the SaaS platform (e.g. Super Admin, Owner, Manager, Staff).
2. **Permission**: Declares individual application access actions (e.g., **create_menu**, **view_reports**).
3. **RolePermission**: M:N bridge mapping permissions to roles.
4. **User**: Operational accounts (owners, kitchen workers, managers). Supports multi-tenant scoping via **restaurant_id**.
5. **UserSession**: Tracks absolute sessions, tokens, IPs, agents, and last activity indexes.
6. **PasswordReset**: Holds temporal opaque hash tokens and expiry markers.

Module B: Restaurant

7. **SubscriptionPlan**: Outlines product tier definitions (billing limits, prices, features).
8. **Restaurant**: The core tenant record defining basic metadata, approval status, and subscription state.
9. **Branch**: Maps geographical restaurant locations for multi-branch environments.
10. **RestaurantImage**: Stores galleries, layouts, and logos with display prioritization tags.
11. **RestaurantSetting**: Schema-less key-value settings for fine-grained tenant preferences.
12. **BusinessHour**: Schedule entries defining opening/closed windows per day of week.

Module C: Menu

13. **Category**: Logical categories for food items (e.g. Appetizers, Desserts, Vegan Special).
14. **FoodItem**: Individual food entries containing descriptions, status, base price details.
15. **FoodImage**: Catalog images for foods highlighting size, color, diet cues.
16. **FoodVariant**: Variations of foods (e.g. Normal size vs Large size base price options).
17. **FoodCustomization**: Extra custom selections (e.g. "Add extra mushrooms", "no onions").

Module D: Nutrition & Ingredients

18. **Nutrition**: Exact macro/micro measurements mapping to a singular food item.
19. **Ingredient**: Global directory of ingredients.
20. **FoodIngredient**: Bridge listing ingredient quantities and quantities inside foods.
21. **Allergen**: System directory of allergens (e.g. peanuts, dairy, gluten).
22. **FoodAllergen**: Red-flag bridge noting allergens in specific food items.
23. **DietaryType**: Category types (e.g. Vegetarian, Keto, Halal, Vegan).
24. **FoodDietaryType**: M:N map of items to dietary types.

Module E: Tables & QR Sessions

25. **RestaurantTable**: Physical restaurant seating details and maximum seat capacity.

- 26. **QrToken**: Cryptographically secure, opaque random tokens mapping to tables.
- 27. **QrSession**: Tracks active table ordering sessions linked to customer devices.
- 28. **TableReservation**: Pre-booked table mappings detailing date, party sizing, status.

Module F: Customers

- 29. **Customer**: Global customer accounts for orders, digital payment defaults, loyalty accounts.
- 30. **CustomerFavorite**: Fast index of customer liked dishes.
- 31. **CustomerRequest**: Live service alerts (e.g. "Water requested", "Call waiter").

Module G: Orders

- 32. **Order**: Core transactional ticket containing totals, status state, linked table and branches.
- 33. **OrderItem**: Line-item details referencing selected variant, quantity, price.
- 34. **OrderStatusHistory**: Step-by-step logs tracking time and staff action changes on orders.

Module H: Kitchen Management

- 35. **KitchenQueue**: Separates orders into specific queues (e.g. Bar, Bakery, Main Kitchen Grill).
- 36. **KitchenOrder**: Routes ordered lines to assigned kitchen queues.
- 37. **PreparationLog**: Micrologs tracking active status timers per dish (e.g., preparation steps).

Module I: Inventory

- 38. **Supplier**: Vendor information for supply chain ordering.
- 39. **InventoryItem**: Holds base tracking info for raw food stock (e.g., milk cartons, potatoes).
- 40. **StockTransaction**: Realtime records of stock changes (e.g. restocks, order deduction, waste).
- 41. **PurchaseOrder**: Purchase transactions with suppliers tracking statuses.

Module J: Payments

- 42. **Payment**: Handles payment state values, methods (Card, Cash, UPI), and references.
- 43. **PaymentTransaction**: Audit log tracking payment gateway responses.

Module K: Marketing & Loyalty

- 44. **Coupon**: Rules for pricing discount codes.
- 45. **CouponUsage**: Logs of unique coupon uses on orders.
- 46. **LoyaltyAccount**: Holds customer reward points balances per restaurant.
- 47. **LoyaltyTransaction**: History ledger of points earned and redeemed.

Module L: Reviews & Communication

- 48. **RestaurantReview**: Customer star ratings and feedback for the tenant.
- 49. **FoodReview**: Specific item level feedback.
- 50. **Feedback**: General site contact forms.

51. **Notification**: User and customer real-time alerts.

52. **NotificationTemplate**: Preset template formats for SMS and push events.

Module M: Reports & Logs

53. **ActivityLog**: Logs critical user interactions.

54. **LoginLog**: Tracks user login counts and IP footprints.

55. **ExpenseRecord**: Tenant ledger for expenses (e.g., rent, power) to yield true analytics.

3. Relationship List

- **SubscriptionPlan to Restaurant:** One-to-Many. Plan details apply to multiple subscribing restaurants.
- **Restaurant to Branch:** One-to-Many. A tenant can operate multiple branches.
- **Restaurant to User:** One-to-Many. A tenant employs multiple users (managers, staff).
- **Role to User:** One-to-Many. A user belongs to one system role.
- **Role to Permission:** Many-to-Many. Mapped via **RolePermission**.
- **Category to FoodItem:** One-to-Many. Categories group multiple menu items.
- **FoodItem to FoodVariant/Customization:** One-to-Many. A menu item supports various sizing and addition setups.
- **FoodItem to Nutrition:** One-to-One. Each food item maps to exactly one nutrition macro profile.
- **FoodItem to Ingredient:** Many-to-Many. Mapped via **FoodIngredient**.
- **FoodItem to Allergen:** Many-to-Many. Mapped via **FoodAllergen**.
- **FoodItem to DietaryType:** Many-to-Many. Mapped via **FoodDietaryType**.
- **RestaurantTable to QrToken:** One-to-Many. A table can receive new QR code entries over time.
- **QrToken to QrSession:** One-to-Many. Each QR token maps to active and historical customer sessions.
- **Customer to Order:** One-to-Many. Customers place multiple orders.
- **Order to OrderItem:** One-to-Many. Orders contain many food items.
- **Order to Payment:** One-to-One. Each order has one final settling payment.
- **Payment to PaymentTransaction:** One-to-Many. Payments can trigger multiple gateway attempts/responses.
- **Supplier to InventoryItem:** One-to-Many. Inventory items can have primary suppliers.
- **InventoryItem to StockTransaction:** One-to-Many. Items track multiple adjustments.

4. Cardinality List

Parent Table	Child Table	Cardinality	Parent Optionality	Child Optionality
SubscriptionPlan	Restaurant	1 : M	Mandatory	Optional (new plan has no users)
Restaurant	Branch	1 : M	Mandatory	Mandatory (min. 1 main branch)
Restaurant	User	1 : M	Optional (Admin role is global)	Optional
Role	User	1 : M	Mandatory	Optional
Branch	RestaurantTable	1 : M	Mandatory	Optional
RestaurantTable	QrToken	1 : M	Mandatory	Mandatory (starts with 1 active)
QrToken	QrSession	1 : M	Mandatory	Optional
Category	FoodItem	1 : M	Mandatory	Optional
FoodItem	Nutrition	1 : 1	Mandatory	Mandatory
Order	OrderItem	1 : M	Mandatory	Mandatory
Order	Payment	1 : 1	Mandatory	Optional (unpaid states)
LoyaltyAccount	LoyaltyTransaction	1 : M	Mandatory	Optional

5. Foreign Key Mapping

All relationships utilize strict database constraints to preserve referential integrity:

Foreign Key Column	Main Table	Referenced Table (Column)	On Delete	On Update
user_id	UserSession	User (id)	CASCADE	CASCADE
restaurant_id	User	Restaurant (id)	RESTRICT	CASCADE
branch_id	RestaurantTable	Branch (id)	RESTRICT	CASCADE
table_id	QrToken	RestaurantTable (id)	CASCADE	CASCADE
active_token_id	QrSession	QrToken (id)	RESTRICT	CASCADE
category_id	FoodItem	Category (id)	RESTRICT	CASCADE
food_item_id	Nutrition	FoodItem (id)	CASCADE	CASCADE
order_id	OrderItem	Order (id)	CASCADE	CASCADE
changed_by	OrderStatusHistory	User (id)	SET NULL	CASCADE
order_id	Payment	Order (id)	RESTRICT	CASCADE
supplier_id	PurchaseOrder	Supplier (id)	RESTRICT	CASCADE

6. Bridge Tables

To satisfy 3rd Normal Form and avoid circular dependencies, all M:N relations are resolved using dedicated bridge tables:

A. table **role_permission**

- **Purpose:** Maps roles to access keys.
- **Keys:** Composite Primary Key (**role_id**, **permission_id**).
- **Indices:** Secondary idx on **permission_id**.

B. table **food_ingredient**

- **Purpose:** Tracks absolute recipe details.
- **Keys:** Composite Primary Key (**food_item_id**, **ingredient_id**).
- **Extra Attributes:** **quantity_weight** (Decimal), **unit** (Varchar).

C. table **food_allergen**

- **Purpose:** Identifies threat allergens for filters.
- **Keys:** Composite Primary Key (**food_item_id**, **allergen_id**).

D. table **food_dietary_type**

- **Purpose:** Powers customer dietary tags (e.g. Vegetarian, Keto).
- **Keys:** Composite Primary Key (**food_item_id**, **dietary_type_id**).

E. table **customer_favorite**

- **Purpose:** Holds customer target catalog favorites.
- **Keys:** Composite Primary Key (**customer_id**, **food_item_id**).

F. table **coupon_usage**

- **Purpose:** Prevents duplicate multi-use code abuse.
- **Keys:** Primary Key (**id**), Unique Composite Key (**coupon_id**, **order_id**, **customer_id**).

7. Database Summary

- **Engine Platform:** MySQL 8.0+
- **Storage Engine:** InnoDB (mandatory for foreign key constraints and ACID transaction compliance).
- **Character Set:** `utf8mb4` with Collation `utf8mb4_unicode_ci` (fully supports emojis, localized content, and case-insensitive searching).
- **Isolation Levels:** Read Committed used for scaling order logs; Repeatable Read for transactions (payment ledger entries).
- **Index Strategy:** Composite indexes on foreign columns (`restaurant_id`, `branch_id`), unique indexing on business keys (slugs, emails, coupon codes), and full-text index on `FoodItem(name, description)`.

8. Validation Notes

Before finalizing, the schema has been checked against these operational criteria:

- **First Normal Form (1NF):** All columns house atomic values. No repeating groups (e.g., multiple phone numbers stored in a single string are normalized out).
- **Second Normal Form (2NF):** Satisfies 1NF and all non-key properties depend entirely on the composite and singular primary keys.
- **Third Normal Form (3NF):** Satisfies 2NF and no transitive functional dependencies remain. (e.g., rather than storing subscription plan amounts directly inside the **Restaurant** table, plans are moved to **SubscriptionPlan** with a reference key).
- **No Orphans Check:** All operational transaction entries map back to high-level tenant templates via constraints.

9. Future Scalability

The schema supports future platform expansions:

- **Multi-Branch & Franchises:** Handled by dividing **Restaurant** (the billing tenant) from **Branch** (the physical restaurant branch processing orders and controlling inventory).
- **Mobile APP APIs:** Fully compatible. The client relies on table QR token sessions (**QrSession**) to order without requiring user accounts.
- **Inventory Forecasting:** Stock transitions are logged as chronological streams (**StockTransaction**), which allows AI tools to compute average usage velocity.
- **Global POS Integrations:** Tables include mapping fields for external identifiers, allowing seamless syncing with systems like Toast POS or Square.